

$$\begin{aligned}
& ((\lambda a. (\lambda b. bb)(\lambda b. bb))b)((\lambda c. (cb))(\lambda a. a)) = \\
& = (\lambda y. \lambda z. ((\lambda x. \lambda y. x) z) (y z)) (\lambda x. \lambda y. x) = \\
& = (\lambda z. ((\lambda x. \lambda y. x) z) ((\lambda x. \lambda y. x) z)) = \\
& = (\lambda z. (\lambda y. z) (\lambda y. z)) = \\
& = (\lambda z. (\lambda y. z) (\lambda y. z)) = \\
& = (\lambda z. z) = I
\end{aligned}$$